

Predicting Hypertension Risk from Lifestyle, Behavioral, and Socioeconomic Factors Using Machine Learning



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Capstone Two

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Executive Summary

This project developed and evaluated machine learning models to predict hypertension using routinely collected demographic, socioeconomic, behavioral, and dietary information from the 2021 Behavioral Risk Factor Surveillance System (BRFSS). The goal was to determine whether commonly available survey data could be used to identify individuals at elevated risk of hypertension and support early intervention without relying on laboratory measurements or electronic health records.

Key Findings

- Developed and compared three machine learning algorithms: Logistic Regression, Decision Tree, and Random Forest.
- Used data from **436,781 U.S. adults** participating in the 2021 BRFSS survey.
- The tuned Random Forest achieved the strongest overall performance with an **accuracy of 72.3%** and **ROC-AUC of 0.789**, while Logistic Regression produced comparable performance with greater interpretability.
- Exploratory analysis demonstrated that hypertension is associated with a combination of demographic, socioeconomic, behavioral, and health-related factors, supporting the use of multidimensional data for risk prediction.
- Findings suggest that machine learning models built using routinely collected survey information could support preventive screening, population health management, and digital health applications by identifying individuals who may benefit from earlier monitoring and lifestyle interventions.

1.0 Introduction

Hypertension, commonly known as high blood pressure, is one of the most significant public health challenges facing the United States today.¹ According to the Centers for Disease Control and Prevention (CDC), nearly half of U.S. adults have hypertension, yet many remain unaware of their condition because symptoms often do not appear until serious health complications have developed². Left untreated, hypertension substantially increases the risk of cardiovascular disease, stroke, kidney disease, and premature death³. Early identification of individuals at increased risk is therefore essential for improving health outcomes and reducing preventable complications.

The burden of hypertension extends beyond individual health. It places considerable strain on healthcare systems through increased hospitalizations, long-term medication use, and the management of chronic diseases associated with uncontrolled blood pressure⁴. Hypertension is also responsible for billions of dollars in annual healthcare expenditures and productivity losses⁵. As healthcare organizations continue to shift toward preventive and value-based models of care, there is growing interest in identifying high-risk individuals before disease progression occurs.

One of the encouraging aspects of hypertension is that many of its risk factors are lifestyle-related and thus, modifiable. Healthy dietary habits, regular physical activity, maintaining a healthy body weight, smoking cessation, and timely access to preventive healthcare have all been shown to reduce the risk of developing hypertension⁶. While some risk factors such as age, sex, and family history cannot be changed, understanding how lifestyle and behavioral factors contribute to hypertension can support targeted prevention strategies and more personalized interventions.

¹ Centers for Disease Control and Prevention. *About High Blood Pressure*. <https://www.cdc.gov/high-blood-pressure/about/index.html>

² Centers for Disease Control and Prevention. *About High Blood Pressure*. <https://www.cdc.gov/high-blood-pressure/about/index.html>

³ American Heart Association. *High Blood Pressure*. <https://www.heart.org/en/health-topics/high-blood-pressure/the-facts-about-high-blood-pressure>

⁴ Goorani S, Zangene S, Imig JD. Hypertension: A Continuing Public Healthcare Issue. *Int J Mol Sci*. 2024 Dec 26;26(1):123. doi: 10.3390/ijms26010123. PMID: 39795981; PMCID: PMC11720251.

⁵ Wang G, Grosse SD, Schooley MW. Conducting Research on the Economics of Hypertension to Improve Cardiovascular Health. *Am J Prev Med*. 2017 Dec;53(6 Suppl 2):S115-S117. doi: 10.1016/j.amepre.2017.08.005. PMID: 29153111; PMCID: PMC5808538.

⁶ Centers for Disease Control and Prevention. *About High Blood Pressure*. <https://www.heart.org/en/health-topics/high-blood-pressure/the-facts-about-high-blood-pressure>

Recent advances in machine learning provide an opportunity to improve population health by identifying complex patterns across large health datasets. By combining demographic, socioeconomic, behavioral, and nutrition-related information, predictive models can estimate an individual's likelihood of hypertension and help healthcare organizations prioritize preventive care efforts. These models are not intended to replace clinical judgment. Rather, they can serve as decision-support tools that assist healthcare providers in identifying patients who may benefit from additional screening, lifestyle counseling, or early intervention.

This project was developed from the perspective of a healthcare technology company interested in using routinely collected health survey information to identify adults at elevated risk of hypertension. Unlike models that rely on laboratory values or electronic health records, this project focuses on information that is relatively easy to collect through health assessments or patient questionnaires. Such an approach has the potential to support scalable population health initiatives, community outreach programs, and digital health applications that encourage preventive care.

The primary objective of this project was to develop and evaluate machine learning models capable of predicting hypertension using demographic, socioeconomic, behavioral, and nutrition-related variables from the 2021 Behavioral Risk Factor Surveillance System (BRFSS). Three classification algorithms were developed and compared, including Logistic Regression, Decision Tree, and Random Forest models. Hyperparameter tuning was performed for the tree-based models to improve generalization and reduce overfitting.

Specifically, this project sought to:

- Develop predictive models to classify individuals with and without hypertension.
- Compare the performance of multiple machine learning algorithms using consistent evaluation metrics.
- Evaluate whether lifestyle and nutrition-related variables provide meaningful predictive value alongside demographic and socioeconomic characteristics.

Through this analysis, the project aims to demonstrate how machine learning can support early risk identification while also highlighting the importance of lifestyle and behavioral factors in chronic disease prevention. The findings are intended to provide practical insights for healthcare technology organizations seeking to build data-driven tools that support preventive healthcare and improve population health outcomes.

2.0 Data

2.1 Data Source

This project uses data from the 2021 Behavioral Risk Factor Surveillance System (BRFSS), an annual health-related telephone survey conducted by the Centers for Disease Control and Prevention (CDC) in collaboration with state health departments. The BRFSS is the largest continuously conducted health survey in the world and collects information on health conditions, health-related risk behaviors, preventive health practices, and access to healthcare among non-institutionalized adults aged 18 years and older in the United States.⁷

The BRFSS was selected because it provides a large, nationally representative dataset containing many of the demographic, behavioral, and lifestyle factors known to be associated with hypertension. Unlike clinical datasets that rely on laboratory measurements or electronic health records, the BRFSS captures information that can be collected through routine health questionnaires. This makes it particularly well suited for developing predictive models that could support population health initiatives, digital health applications, and preventive screening programs.

For this project, data from the 2021 survey cycle were used because it included data on all the variables needed for the analysis. After data preparation, the final analytical dataset contained **436,781 respondents**, providing sufficient statistical power for both exploratory analysis and machine learning model development.

Table 1. BRFSS Dataset Summary⁸

Characteristic	Description
Data source	Behavioral Risk Factor Surveillance System (BRFSS)
Organization	Centers for Disease Control and Prevention (CDC)
Survey year	2021
Study population	Non-institutionalized U.S. adults aged 18 years and older
Sample size	436,781 respondents
Target variable	Hypertension (self-reported diagnosis)

⁷ Centers for Disease Control and Prevention. *Behavioral Risk Factor Surveillance System (BRFSS): Overview*. <https://www.cdc.gov/brfss/>

⁸ Centers for Disease Control and Prevention. *2021 BRFSS Survey Data and Documentation*. https://www.cdc.gov/brfss/annual_data/annual_2021.html

2.2 Outcome Variable

The objective of this project was to predict whether a respondent had previously been diagnosed with hypertension. The target variable was derived from the BRFSS calculated hypertension indicator, which classifies respondents according to whether they have ever been told by a healthcare professional that they have high blood pressure. For machine learning purposes, the outcome variable was recoded into a binary classification, where 0 represented respondents without hypertension and 1 represented respondents with hypertension.

2.3 Predictor Variables

Twenty predictor variables were selected based on existing evidence linking demographic, socioeconomic, behavioral, lifestyle, and nutritional factors with hypertension risk, as well as their relevance to population health screening. Preference was given to variables that are routinely collected during health assessments, allowing the resulting models to remain practical for real-world implementation.

Rather than including every available BRFSS variable, a focused subset of predictors was selected to balance predictive performance with interpretability. Variables were grouped into six broad categories.

Table 2. Predictor Variables Included in the Analysis⁹

Category	Representative variables
Demographic	Age, Sex, Race/Ethnicity, Marital Status
Socioeconomic	Education Level, Employment Status, Income Group
Healthcare Access	Health Insurance Coverage, Personal Healthcare Provider
Health & Lifestyle	Body Mass Index (BMI), Physical Activity, Smoking Status, Alcohol consumption, E-cigarette Use, Physical Health Status
Nutrition	Total Fruit Consumption, Total Vegetable Consumption, French Fry Consumption
Geographic	Metropolitan Status, Urban/Rural Classification

Several additional variables available within the BRFSS dataset were evaluated during the exploratory data analysis but were excluded from the final models to reduce redundancy, improve interpretability, and minimize multicollinearity. For example, dark green vegetable consumption was removed because it was highly correlated with overall vegetable

⁹ Centers for Disease Control and Prevention. 2021 BRFSS Codebook.
https://www.cdc.gov/brfss/annual_data/annual_2021.html

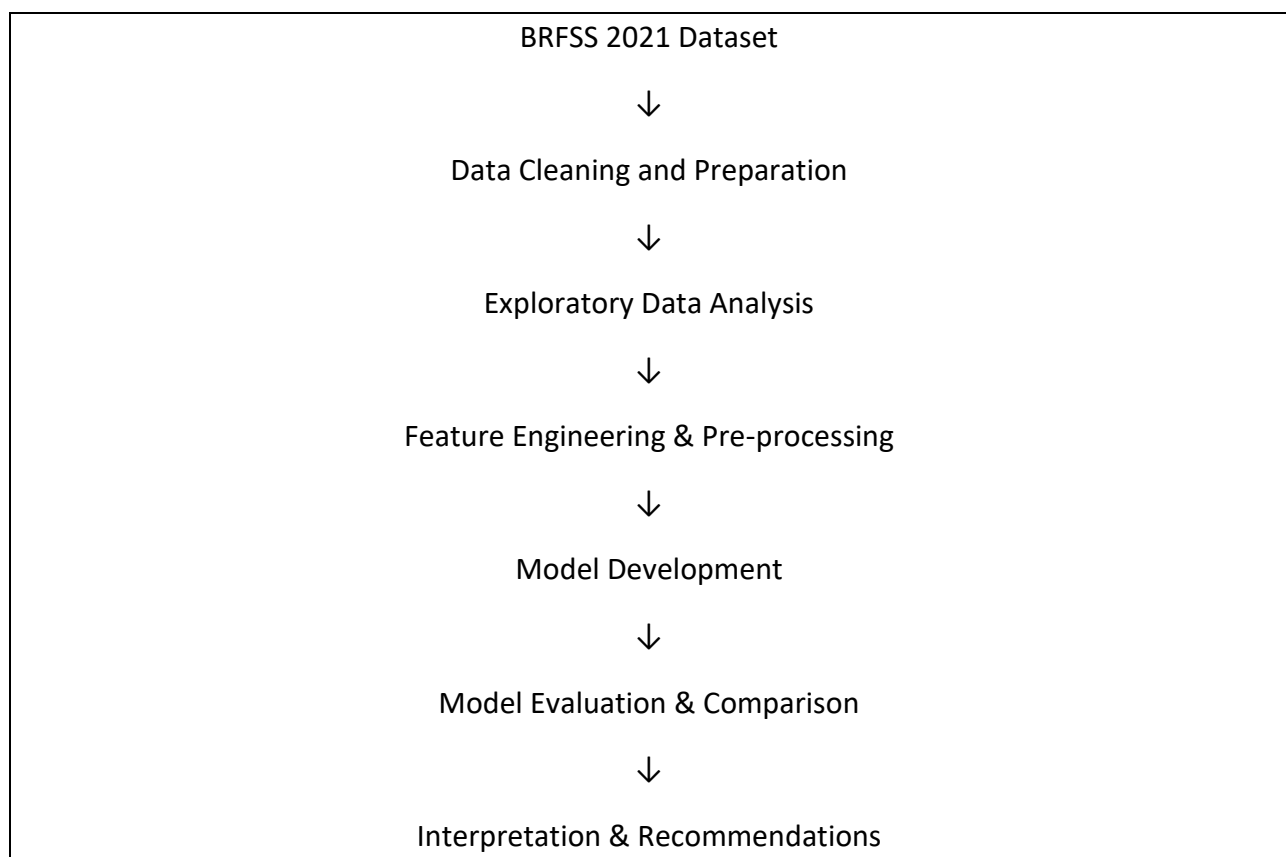
consumption, while alcohol consumption was excluded due to concerns regarding the reliability of self-reported intake.

The final dataset provides a comprehensive representation of demographic, behavioral, socioeconomic, and nutrition-related factors associated with hypertension risk while remaining suitable for predictive modeling and interpretation.

3.0 Methodology

This project followed a structured machine learning workflow, beginning with raw survey data and progressing through data preparation, exploratory data analysis (EDA), predictive modeling, and interpretation of results. The objective was to develop an accurate predictive model for hypertension while evaluating the predictive value of demographic, behavioral, socioeconomic, and nutrition-related factors. Figure 1 summarizes the overall analytical workflow used throughout the project.

Figure 1. Analytical Workflow



4.0 Data Cleaning and Preparation

The analysis began by selecting a subset of variables from the 2021 BRFSS dataset that were relevant to hypertension prediction. Survey-specific response codes representing "Don't know," "Refused," and other missing responses were converted to missing values (NaN) to ensure they were handled consistently throughout the analysis.

Several variables required additional cleaning before modeling. Exploratory analysis identified implausible values within the BMI and nutrition variables that were inconsistent with realistic physiological or dietary behavior. BMI values outside a plausible adult range were treated as missing observations. The nutrition variables also contained extremely small artifact values, which were converted to zero, along with unusually large consumption values that were unlikely to represent realistic dietary patterns. To reduce the influence of these observations, upper thresholds were established for fruit, vegetable, dark green vegetable, and French fry consumption based on plausible daily intake, with observations exceeding these thresholds converted to missing values prior to imputation (Table 3).

Table 3. Plausible Daily Intake Thresholds Applied During Data Cleaning

Variable	Maximum Daily Intake
Total fruit consumption	10 servings/day
Total vegetable consumption	15 servings/day
Dark green vegetable consumption	5 servings/day
French fry consumption	3 servings/day

Missing data were evaluated for each variable before selecting an appropriate treatment strategy. Variables with less than 2% missing data were imputed using the mode, while variables with between 2% and 10% missing data were imputed using the mode and supplemented with binary missing indicator variables. Variables with between 10% and 20% missing data were imputed using the median, together with corresponding missing indicator variables. Income was the only variable with more than 20% missing observations. Rather than imputing these values, respondents were grouped into low, medium, and high-income categories, with missing income retained as a separate category. This approach acknowledged that income may not be missing at random and preserved potentially meaningful socioeconomic information for the predictive models (Table 4).

Table 4. Dealing with Missing Data

Missing Data Category	Treatment
<2%	Mode imputation
2–10%	Mode + missing indicator
10–20%	Median + missing indicator
>20%	Categorize variable and retain missing as separate category

These preprocessing steps produced a clean analytical dataset while retaining the maximum number of observations for subsequent exploratory analysis and machine learning.

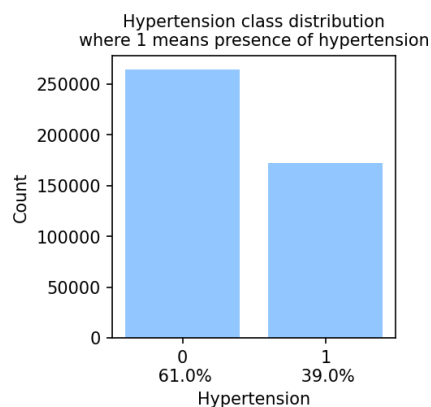
5.0 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to better understand the distribution of each variable, identify missing data patterns, detect outliers, and examine relationships between predictor variables and hypertension. Visualizations and summary statistics were used to compare respondents with and without hypertension and to identify variables that demonstrated meaningful associations with the outcome.

5.1 Hypertension Prevalence

Approximately **39.4%** of respondents in the 2021 BRFSS dataset reported having been diagnosed with hypertension, while **60.6%** reported no diagnosis. Although the classes are not perfectly balanced, the distribution is sufficiently even that extensive resampling techniques were not required during model development. Stratified train-test splitting was used to preserve this class distribution throughout model training and evaluation.

Figure 2. Hypertension Prevalence



5.2 Summary Statistics

Table 5 summarizes the continuous variables included in the analysis. The average respondent was approximately 55 years old with a mean BMI of 28.4 kg/m², indicating that, on average, respondents fell within the overweight BMI category. Average Fruit and Vegetable consumption was approximately 1.3 and 1.9 equivalent servings per day, respectively, while French fry consumption remained relatively low. These descriptive statistics provide an overview of the study population before examining relationships with hypertension.

Table 5. Summary Statistics of Numerical Features

Variable	Mean	SD
Age	54.7	17.6
BMI	28.5	6.1
Physical Health Days	1.4	0.7
Fruit Consumption	1.3	1.1
Vegetable Consumption	1.9	1.3
French Fry Consumption	0.2	0.3

5.3 Relationships with Hypertension

Several interesting patterns emerged during the exploratory analysis. Hypertension prevalence increased steadily with age, with respondents aged 65 years and older exhibiting the highest prevalence (Figure 3). A similar trend was observed for BMI, where respondents in higher BMI categories were considerably more likely to report hypertension (Figure 4). Nutrition variables showed consistent associations, with respondents consuming fewer fruits and vegetables generally exhibiting higher hypertension prevalence (Figure 5). These findings are consistent with established risk factors reported in the public health literature and support the inclusion of these variables in the predictive models.

Figure 3. Age vs Hypertension

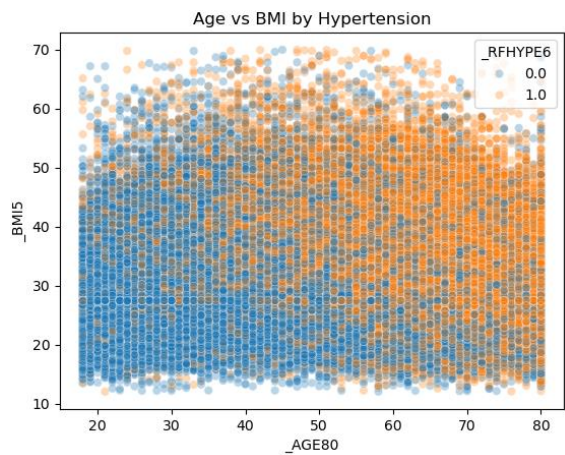


Figure 4. BMI vs Hypertension

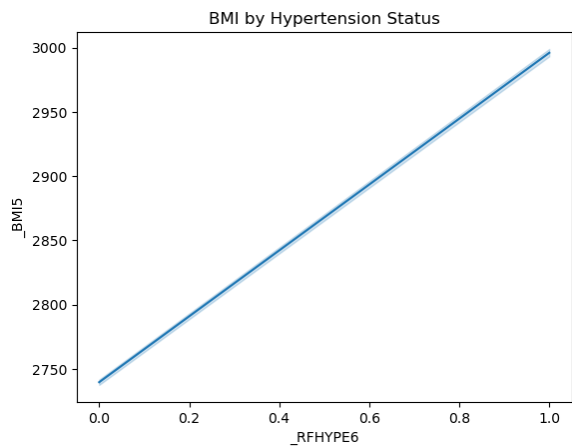
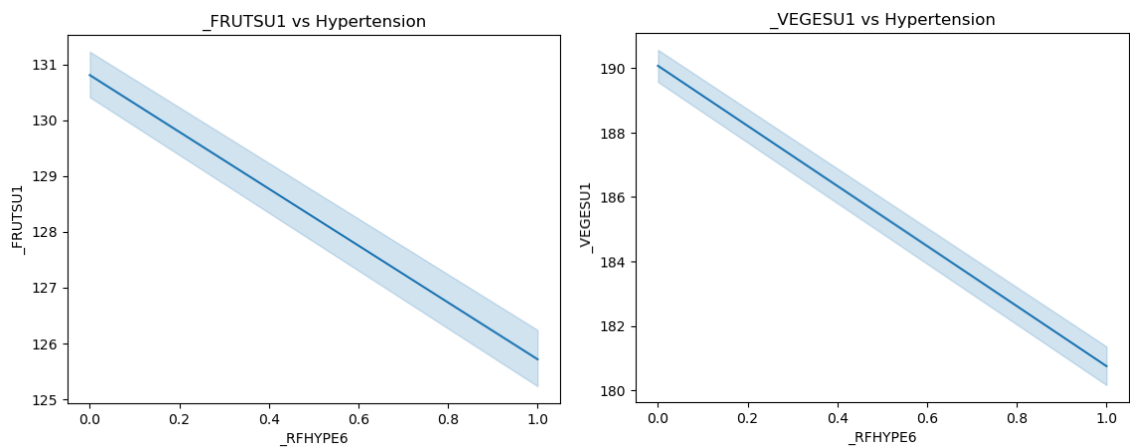


Figure 5. Fruit and Vegetable consumption vs Hypertension



5.4 Key Insights from EDA

Beyond the relationships observed for age, BMI, and dietary factors, the exploratory analysis highlighted several additional patterns associated with hypertension. Socioeconomic characteristics appeared to play an important role. Respondents in the lowest income category reported the highest prevalence of hypertension, while those in the highest income category had the lowest prevalence. Employment status showed a similar trend, with retired respondents and those unable to work experiencing substantially higher rates of hypertension than respondents who were employed or students. These findings suggest that socioeconomic circumstances may influence hypertension risk, either directly or through differences in healthcare access, lifestyle, and underlying health conditions.

Several behavioral factors also demonstrated meaningful relationships with hypertension. Respondents who reported no physical activity during the previous 30 days had a noticeably higher prevalence of hypertension than those who were physically active. Former smokers exhibited the highest prevalence of hypertension among the smoking categories, while respondents reporting fair or poor overall health, or a greater number of physically unhealthy days, were also more likely to report hypertension. Although alcohol consumption appeared to be associated with hypertension, the observed pattern was difficult to interpret and may reflect reporting bias inherent in self-reported survey data. Consequently, alcohol consumption was not included in the final predictive models.

The exploratory analysis also informed several feature selection decisions. Correlation analysis showed generally weak relationships among the continuous predictor variables, suggesting that multicollinearity was unlikely to substantially affect model performance. The only notable exception was the moderate correlation between total vegetable consumption and dark green vegetable consumption. Because these variables captured similar dietary behavior, dark green vegetable consumption was excluded from the final models to reduce redundancy while maintaining interpretability.

Overall, the exploratory analysis demonstrated that hypertension is associated with a combination of demographic, socioeconomic, behavioral, and health-related factors rather than any single characteristic. These findings supported the selection of predictors for the machine learning models and provided confidence that the final models would capture multiple dimensions of hypertension risk.

6.0 Feature Engineering and Pre-processing

Categorical variables were converted into indicator (dummy) variables to enable their use in machine learning algorithms. Continuous variables were standardized using the StandardScaler

to place predictors on comparable scales where appropriate. The target variable was recoded into a binary outcome indicating the presence or absence of hypertension.

The processed dataset was then divided into training (80%) and testing (20%) subsets using stratified sampling to preserve the proportion of respondents with hypertension in both datasets. All preprocessing decisions were performed using only the training data where appropriate to avoid data leakage.

7.0 Model Development

Three supervised classification algorithms were selected for evaluation: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was chosen as an interpretable baseline model, while the tree-based algorithms were selected because of their ability to capture nonlinear relationships and interactions among predictor variables.

Baseline models were developed using default model parameters before applying hyperparameter tuning to the Decision Tree and Random Forest algorithms. GridSearchCV was used to optimize the Decision Tree model, while RandomizedSearchCV was used for the Random Forest model to efficiently explore the larger hyperparameter space while reducing computational time.

Model performance was evaluated using an independent testing dataset that was not used during model training. Performance was assessed using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). These metrics were selected because they provide complementary information regarding overall predictive performance, the balance between false positives and false negatives, and each model's ability to discriminate between respondents with and without hypertension. Attention was given to recall (sensitivity), which measures the proportion of individuals with hypertension who were correctly identified by the model. In the context of preventive healthcare, minimizing false negatives is especially important because failing to identify high-risk individuals may delay screening, lifestyle interventions, and appropriate clinical follow-up.

Comparing multiple evaluation metrics allowed for a more comprehensive assessment of model performance and supported the selection of the final model used for interpretation and recommendations.

7.1 Logistic Regression

Logistic Regression was selected as the baseline model because it is widely used for binary classification problems and provides an interpretable benchmark against which more complex algorithms can be compared.

The model achieved an accuracy of **72.0%**, a recall of **59.0%** and a ROC-AUC of **0.786**, demonstrating strong overall performance and balanced precision and recall. These results indicate that the selected predictors explain a substantial proportion of hypertension risk using a relatively simple linear model.

Table 6. Logistic Regression Performance

Metric	Value
Accuracy	0.720
Precision	0.662
Recall	0.590
F1 Score	0.624
ROC-AUC	0.786

7.2 Decision Tree (Baseline and Tuned)

Decision Trees were evaluated because they can capture nonlinear relationships and interactions between predictor variables while remaining relatively easy to interpret.

The baseline Decision Tree substantially overfit the training data, achieving near-perfect training accuracy (99%) but much poorer performance on the testing dataset (63%).

Hyperparameter tuning using GridSearchCV reduced overfitting considerably and improved the model's ability to generalize to unseen data. The tuned model performed substantially better than the baseline Decision Tree, with a **72.0%** accuracy, **61.4%** recall, and a ROC-AUC score of **0.779**, but overall performance remained slightly below the Logistic Regression model.

Table 7. Decision Tree Performance (Baseline vs Tuned)

Metric	Baseline	Tuned
Training Accuracy	0.999	0.772
Testing Accuracy	0.632	0.716
Precision	0.533	0.648
Recall	0.538	0.614
F1 Score	0.535	0.630
ROC-AUC	0.615	0.779

7.3 Random Forest (Baseline and Tuned)

Random Forest was selected because it combines predictions from multiple decision trees, resulting in more stable and accurate predictions while reducing overfitting.

The baseline Random Forest produced strong predictive performance but exhibited evidence of overfitting (99% training accuracy to 72% in testing). Hyperparameter tuning using RandomizedSearchCV substantially reduced the gap between training and testing accuracy while slightly improving predictive performance. The tuned Random Forest achieved the highest overall accuracy (**72.3%**), Recall (**60.9%**) and ROC-AUC (**0.789**), making it the strongest-performing model evaluated in this study.

Table 8. Random Forest Performance (Baseline vs Tuned)

Metric	Baseline	Tuned
Training Accuracy	0.999	0.760
Testing Accuracy	0.715	0.723
Precision	0.650	0.661
Recall	0.597	0.609
F1 Score	0.623	0.634
ROC-AUC	0.779	0.789

7.4 Model Comparison

Given that hypertension prediction requires balancing sensitivity and specificity, Accuracy, Recall, and ROC-AUC were considered key metrics for comparing model performance. Table 9 compares the performance of all models evaluated in this study. Logistic Regression provided a strong and highly interpretable baseline, achieving the second highest Accuracy (72.0%) and ROC-AUC (0.786) while Recall was slightly lower than the other models (59.0%). Both tree-based algorithms benefited substantially from hyperparameter tuning, which reduced overfitting and improved generalization. Among all models, the tuned Random Forest achieved the best overall balance between accuracy, recall, precision, F1 score, and ROC-AUC and was therefore selected as the final model for discussion and recommendations.

Table 9. Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic regression	0.720	0.662	0.590	0.624	0.786
Baseline Decision Tree	0.632	0.533	0.538	0.535	0.615
Tuned Decision Tree	0.716	0.648	0.614	0.630	0.779
Baseline Random Forest	0.715	0.650	0.597	0.623	0.779
Tuned Random Forest	0.723	0.661	0.609	0.634	0.789

8.0 Discussion and Business Implications

8.1 Key Findings

This project demonstrates that machine learning can effectively predict hypertension using routinely collected demographic, socioeconomic, behavioral, and nutrition-related information. Across the models evaluated, the results showed that these variables collectively provide meaningful predictive information for identifying individuals at elevated risk of hypertension. The findings highlight the value of combining multiple dimensions of health and lifestyle data to support hypertension risk assessment and reinforce the potential of machine learning as a tool for early identification of high-risk individuals.

Among the models evaluated, Logistic Regression provided a strong and highly interpretable baseline, while the tuned Random Forest achieved the best overall predictive performance. Although the improvement over Logistic Regression was modest, hyperparameter tuning substantially reduced overfitting in the tree-based models and improved their ability to generalize to unseen data. This suggests that the relationships between the selected predictors and hypertension are largely captured by simple linear associations, with only modest gains achieved by modeling more complex interactions.

The exploratory analysis further demonstrated that hypertension is associated with a combination of demographic, socioeconomic, behavioral, and health-related factors rather than any single characteristic. These findings reinforce the importance of considering multiple dimensions of health when developing predictive models for chronic disease risk.

8.2 Business Implications

From a healthcare technology perspective, the findings suggest that routinely collected survey or questionnaire data can be used to identify individuals at elevated risk of hypertension without relying on laboratory measurements or electronic health records. This creates opportunities for scalable risk stratification within digital health platforms, employer wellness programs, community health initiatives, and population health management programs.

The strong performance of the Logistic Regression model also highlights an important practical consideration. While the tuned Random Forest achieved the highest overall predictive performance, the improvement over Logistic Regression was relatively small. Organizations that prioritize transparency and model interpretability may therefore prefer Logistic Regression, particularly in clinical settings where understanding the contribution of individual risk factors is important. Conversely, organizations seeking to maximize predictive performance may favor the tuned Random Forest, accepting a modest reduction in interpretability in exchange for improved accuracy and recall.

Overall, the results demonstrate that machine learning models can support, rather than replace, clinical decision-making by helping healthcare providers identify individuals who may benefit from additional screening, preventive counseling, or targeted lifestyle interventions.

8.3 Study Limitations

Several limitations should be considered when interpreting the results of this study. First, the BRFSS relies on self-reported information, which may be affected by recall bias and social desirability bias. Variables such as alcohol consumption, smoking, physical activity, and dietary intake are particularly susceptible to inaccurate reporting.

Second, the BRFSS is a cross-sectional survey that captures information at a single point in time. Consequently, while the models identify associations between predictor variables and hypertension, they cannot establish causal relationships.

Third, this analysis was limited to data collected during the 2021 BRFSS survey. Although the dataset is large and nationally representative, incorporating multiple survey years may improve model robustness and better capture temporal changes in population health and health behaviors.

Finally, the models were developed using survey data rather than clinical measurements or longitudinal patient records. As a result, the findings are most applicable to population-level risk screening and should not be interpreted as a substitute for clinical diagnosis or individualized medical assessment.

9. Recommendations

Based on the findings of this study, healthcare organizations should consider incorporating machine learning models into preventive screening programs to identify individuals at elevated risk of hypertension before clinical diagnosis. Because the model relies on routinely collected demographic, socioeconomic, behavioral, and health-related information, it could be integrated into electronic health records or digital health platforms without requiring additional laboratory testing. Individuals identified as high risk could then be prioritized for blood pressure monitoring, lifestyle counseling, or referral to preventive care programs.

The results also highlight the importance of collecting comprehensive patient information beyond traditional clinical measures. Demographic, lifestyle, and socioeconomic factors together provide valuable information for predicting hypertension risk, suggesting that a holistic approach to data collection may improve early risk identification. While the Random Forest model achieved the strongest predictive performance, organizations may also consider

Logistic Regression in situations where model transparency and ease of interpretation are priorities.

10. Future Work

Several opportunities exist to further improve this model and expand its potential applications:

- **Expand the training dataset:** Incorporate multiple years of BRFSS data to increase the size and diversity of the training dataset and evaluate whether model performance remains consistent over time.
- **Include additional clinical variables:** Incorporate variables such as laboratory values, medication history, or family history of hypertension to improve predictive accuracy beyond what is possible using survey data alone.
- **Evaluate feature importance:** Identify the variables that contribute most to hypertension prediction. This information could be used to streamline data collection by prioritizing the most informative survey questions, making the model more practical for implementation in settings with time or resource constraints.
- **Explore additional machine learning approaches:** Evaluate more advanced machine learning techniques to determine whether further improvements in predictive performance can be achieved.
- **Conduct external validation:** Test the model using data from other population-based surveys or healthcare systems to assess its generalizability and suitability for real-world implementation.